



COLE MCGREGOR



Contact

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Education

Siena College

*B.S. in Software Development
Software Engineering Specialization
Cum Laude, 2025*

Hudson Valley Community College

*A.S. in Computer Science, 3.7 GPA,
2022*

Technical Skills

Languages

- C++
- Java
- JavaScript
- Python
- C
- C#

Cloud & Infrastructure

- AWS
- Google Cloud
- Docker
- Git/GitHub

Personal Profile

Software systems engineer and owner of McGregor Systems LLC, specializing in software development, systems integration, workflow automation, cloud-based solutions, and operational software improvement. Experienced with Java, C#, JavaScript, cloud tooling, and API-driven systems. Former United States Air Force Armament Systems Specialist with experience in regulated technical environments requiring precision, accountability, and procedural compliance.

Relevant Experience

SOFTWARE SYSTEMS ENGINEER

RAS Systems | McGregor Systems LLC, '25 - '26

- Stabilized reporting workflows spanning JavaScript, AWS/Bubble storage, and third-party integrations
- Refactored brittle logic and templates to improve reliability and maintainability
- Implemented admin access controls and workflow improvements
- Rebuilt multi-page PDF generation logic with improved layout handling
- Documented system behavior and provided architectural recommendations

PRODUCT OWNER | SOFTWARE DEVELOPER

AI Healthcare Scheduler, Software Engineering Capstone, '24 - '25

- Led development of a service-oriented scheduling system using FastAPI, Docker, and AWS
- Designed backend services for authentication, logging, and integrations
- Integrated external APIs including Twilio and OpenAI
- Worked with AWS and Google Cloud resources to support deployment and integration
- Produced architecture documentation and coordinated Agile workflows

Systems & Development

- API Integration
- Workflow Automation
- Refactoring
- Software Modernization
- Database Design
- Technical Documentation

Frameworks & Platforms

- Spring Boot
- Node.js
- React Native
- Firebase

Game Engines

- Unreal
- Unity
- Godot



Core Competencies

- Workflow Automation
- Software Modernization
- Legacy Refactoring
- Systems Troubleshooting
- API Integration
- Cloud Deployment
- Technical Documentation
- Requirements Analysis
- Client Communication
- Operational Support
- Process Improvement

ANALYTICS & STATISTICAL TRACKING DEVELOPER

Collaborative Fitness Mobile Application Project, '25

- Designed and implemented statistical tracking and analytics functionality for a collaborative fitness application using Firebase-backed data storage
- Developed multi-vector graph representations of user workout and progression data to support trend analysis and performance visualization
- Built data aggregation and reporting systems to provide actionable user performance insights across multiple exercise metrics
- Worked with application data structures and backend logic to improve reporting accuracy, maintainability, and visualization workflows

SOFTWARE SYSTEMS & SIMULATION DEVELOPER

Independent Development Project, '26

- Developed modular simulation architecture in C++ supporting procedural generation, entity management, and state-driven systems
- Built visualization and testing harnesses to validate runtime behavior, rendering logic, and generation systems
- Implemented pathfinding, simulation control, and extensible world-state management systems
- Designed tooling and debugging workflows to support iterative development and system validation

ARMAMENT SYSTEMS SPECIALIST

United States Air Force, '16-'19

- Performed maintenance, inspections, and troubleshooting on F-15E aircraft armament and weapons systems in high-tempo flightline environments
- Managed munitions accountability, inventory tracking, inspections, and technical documentation under strict procedural and safety standards
- Supervised personnel, training, task coordination, and procedural compliance within support squadron and maintenance operations
- Coordinated with maintenance and flightline teams to support aircraft generation, operational readiness, and mission requirements
- Operated in highly regulated, security-sensitive technical environments requiring precision, accountability, and disciplined execution